GLOSSARY

Round
One year of play. You invest first, then trade through four seasons.

Season
One trading period. You set your prices, then the market simulates one week and multiplies the result by 13.

Merit order
All offers sorted from cheapest to most expensive. The cheapest are used first.

Clearing price
The price where supply meets demand. Every offer that sells is paid this same price.

Must-serve demand
Demand that has to be met at any price. If it is not, there is a blackout.

Capacity factor
A facility's average output as a share of its maximum power.

Marginal cost
What it costs to produce one more MWh.

LCOE
Levelized cost of electricity: everything a facility costs over its life, divided by the energy it produces.

Dumping
Renewable output that does not sell is thrown away, at a cost of 25 ¤/MWh.

1 Round = 1 year = 364 days = 8,736 h · 1 season = 13 weeks = 2,184 h · Prices are per MWh.



Your score

Score = operating profit − investments + residual value

The Scoreboard page shows the full breakdown.

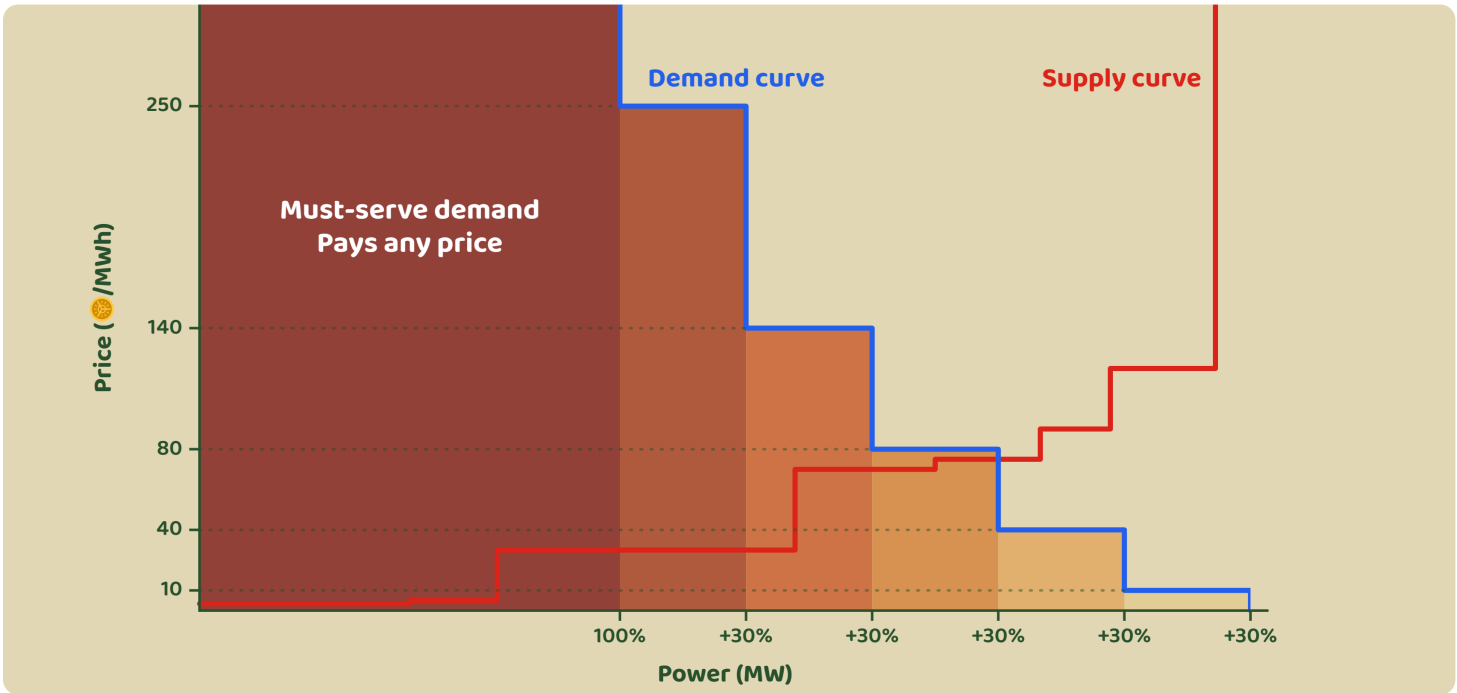
⚠ Blackout

If must-serve demand cannot be met, the grid goes down for a day. Demand is then cut by 10% for the rest of the Round, and each blackout costs **every player** 30% of their operating profit. Keep an eye on the control room screen.

⚠ Events

High emission levels make events with negative consequences more likely. The Round Recap warns you about them: read it carefully.

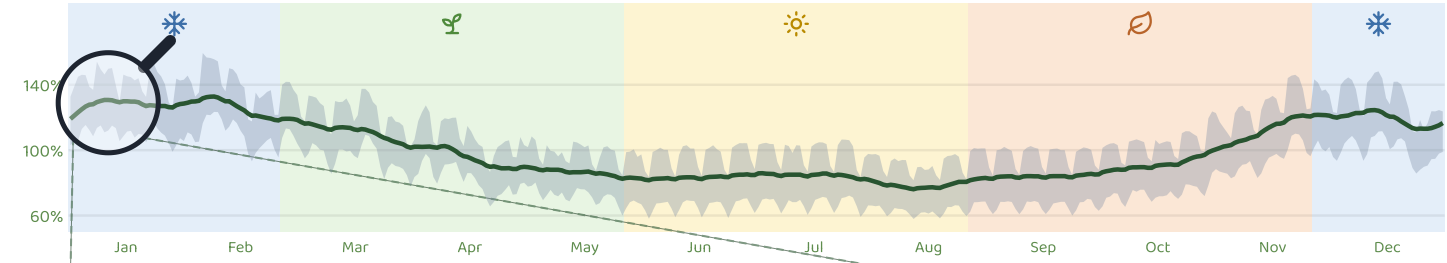
THE SIX DEMAND BLOCKS



THE YEAR AT A GLANCE

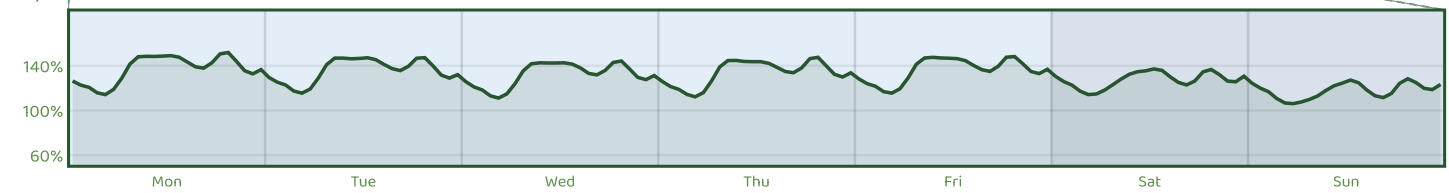
Demand over the year

% of the yearly average · line: smoothed · band: lowest to highest of each day

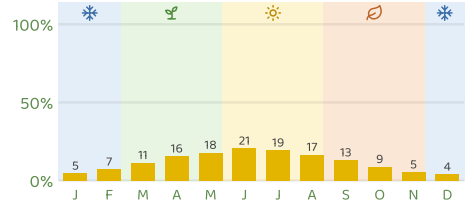


Zoom: the winter week the market simulates

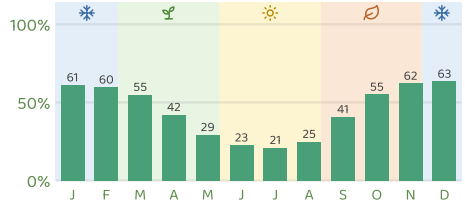
Every season simulates its own middle week · weekends shaded



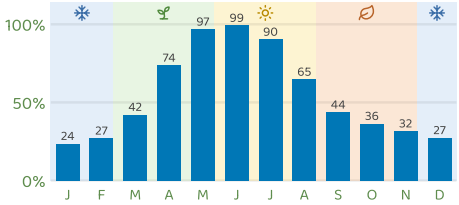
Solar



Wind

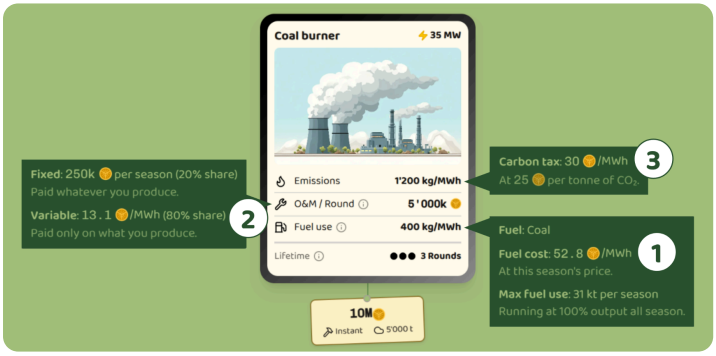


Hydro



Bars: average output as a share of maximum power (capacity factor), per month.

MARGINAL COST



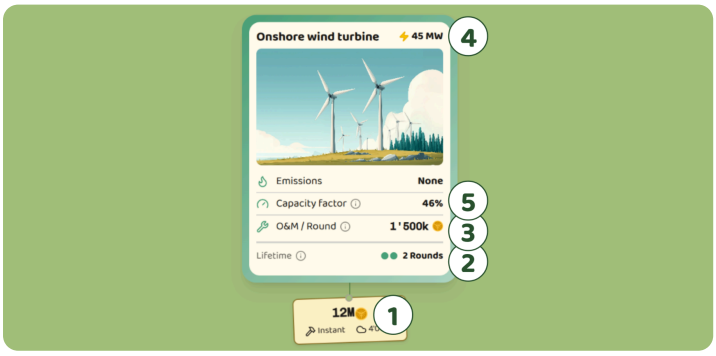
Marginal cost [/MWh] = ① + ② + ③

① Fuel cost = fuel price × fuel use
② Variable O&M
③ Carbon tax cost = carbon tax per kg CO_2 × emissions

Example, coal burner: 52.8 + 13.1 + 30 = **95.9 /MWh**

What is the marginal cost of a renewable power plant?

LCOE



LCOE [/MWh] = $\dfrac{① + ② \times ③}{④ \times 8{,}736\ \text{h} \times ⑤ \times ②}$

① Build price
② Lifetime in Rounds
③ Operational cost per Round = O&M + fuel cost
④ Installed power capacity
⑤ Capacity factor

Example, onshore wind turbine:

$$\frac{12\text{M} + 2 \times 1.5\text{M}}{45\ \text{MW} \times 8{,}736\ \text{h} \times 46\% \times 2} = \textbf{41.5 /MWh}$$

THINK ABOUT IT

Why is LCOE not equal to marginal cost?

What is the optimal bidding price for renewables?

Which one is most useful to help you set your prices?

What happens if everyone bids high, e.g. 900 per MWh?